

EKLAKH DEWAN

AI Engineer — RAG — Agentic AI — Retrieval Engineering

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SUMMARY

AI Engineer focused on retrieval-augmented generation, agentic workflows, and backend AI systems. Experienced in dense and sparse retrieval, hybrid search, cross-encoder reranking, evidence validation, LLM API integration, and evaluation-driven development using Python and FastAPI.

EDUCATION

KPRIET

B.Tech. in Artificial Intelligence & Data Science

2027

CGPA: 8.55

EXPERIENCE

Artificial Intelligence Intern

Flowrage Technology

Virtual Internship — 4 Weeks

- Integrated the **Google Gemini API** into AI-driven application workflows and developed components for automated resume analysis and job recommendation.
- Designed NLP processing pipelines to transform unstructured resumes and job descriptions into structured representations for downstream matching.
- Applied **TF-IDF** and dense semantic embeddings to capture both lexical and semantic similarity between candidate profiles and job requirements.
- Developed and refined resume-matching logic, connecting text preprocessing, feature extraction, similarity scoring, and recommendation workflows.

SELECTED PROJECTS

Enterprise RAG / AI Search Platform

[GitHub](#)

Python, FastAPI, FAISS, BM25, SentenceTransformers, React

- Engineered dense **FAISS** and sparse **BM25** retrieval with hybrid fusion and cross-encoder reranking for evidence retrieval.
- Implemented evidence-constrained generation, citation mapping, and faithfulness validation to ground generated responses in retrieved context.
- Built evaluation workflows using **Recall@K**, **MRR**, and **nDCG** to assess retrieval quality across pipeline stages.

APX — Accounts Payable Exception Resolution Agent

[GitHub](#)

Python, BM25, Dense Retrieval, Reranking, LLMs

- Built an evidence-driven exception-resolution pipeline spanning deterministic validation, retrieval, agent, decision, and persistence stages.
- Combined **BM25**, dense retrieval, hybrid fusion, cross-encoder reranking, and evidence-validity checks before resolution decisions.
- Designed deterministic guardrails and persistence boundaries for controlled and auditable AI-assisted workflows.

TaxTrace — AI-Assisted Tax Reconciliation Platform

[GitHub](#)

FastAPI, Next.js, PostgreSQL, SQLAlchemy, Alembic

- Developed an AI-assisted tax reconciliation platform supporting explainable exception review and structured notice-response workflows.
- Built a Next.js review workspace backed by **FastAPI**, **PostgreSQL**, **SQLAlchemy**, **React Query**, and **Alembic** migrations.
- Added task, exception, notice-case, and draft persistence; Stage 6 backend validation reached **91 passing tests**.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript

AI / ML: RAG, LLM APIs, Agentic Workflows, NLP, Embeddings, SentenceTransformers, scikit-learn

Retrieval: FAISS, BM25, Hybrid Search, RRF, Cross-Encoder Reranking, Semantic Search

Backend: FastAPI, REST APIs, PostgreSQL, SQLAlchemy

Infrastructure: Docker, GitHub Actions, CI/CD, Cloudflare Workers

Frontend: React, Next.js, Vite, Streamlit, HTML/CSS

ENGINEERING FOCUS

Evidence-grounded generation — Retrieval evaluation — AI pipeline architecture — Guardrails — Observability — API integration — Automated testing